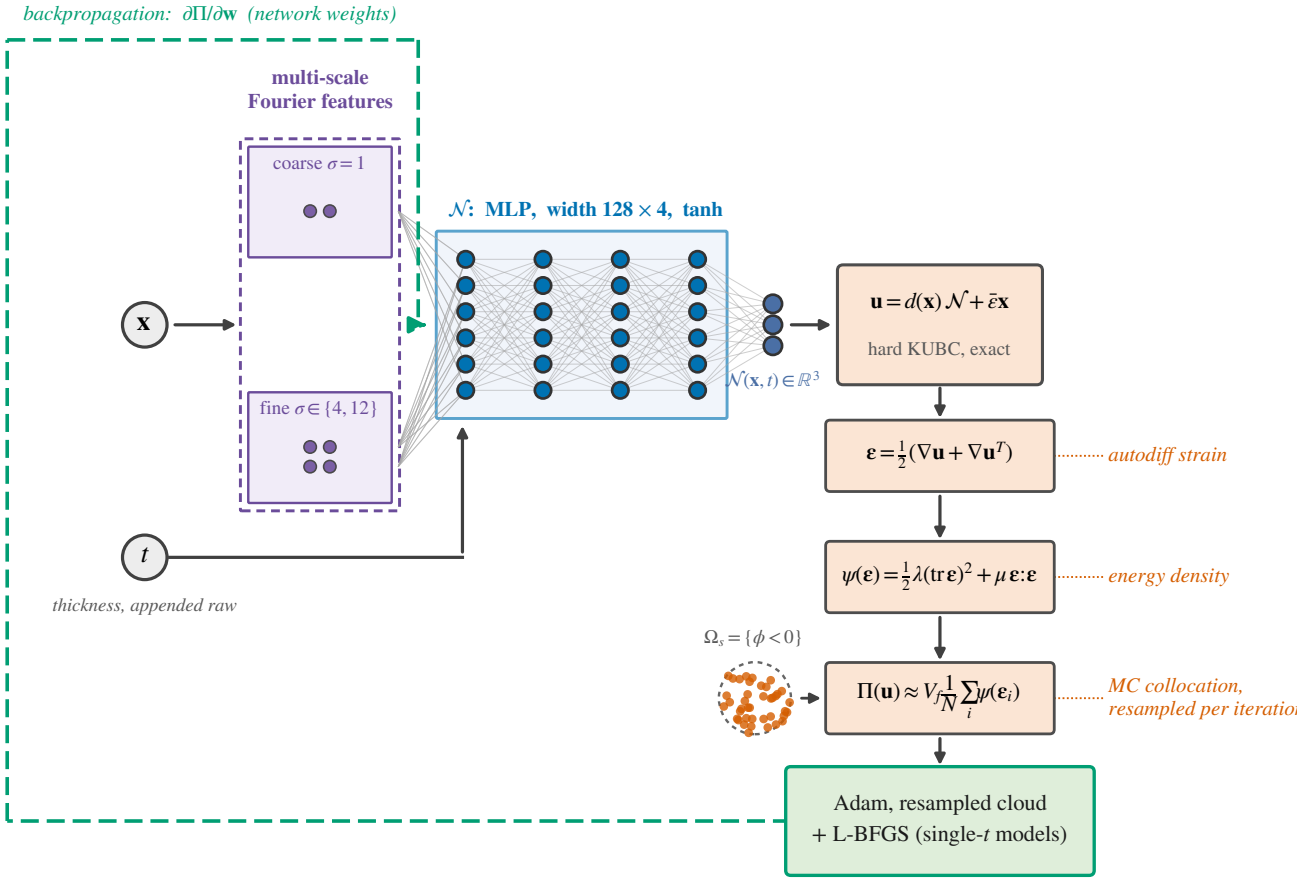
backpropagation: ∂Π/∂w (network weights)
multi-scale
Fourier features
coarse σ = 1
fine σ ∈ {4, 12}
x
t
thickness, appended raw
𝒩: MLP, width 128 × 4, tanh
𝒩(x, t) ∈ ℝ³
u = d(x) 𝒩 + ε̄x
hard KUBC, exact
ε = ½(∇u + ∇uᵀ)
autodiff strain
ψ(ε) = ½λ(tr ε)² + μ ε:ε
energy density
Ω_s = {φ < 0}
Π(u) ≈ V_f (1/N) Σ_i ψ(ε_i)
MC collocation,
resampled per iteration
Adam, resampled cloud
+ L-BFGS (single-t models)